

Daria Schumm

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EDUCATION

Doctor of Philosophy

July 2023 – July 2027

University of Zürich

Zürich, Switzerland

- Thesis: Evaluation Framework for Decentralized Identity Systems
- Supervisor: Prof. Dr. Burkhard Stiller

Master of Information Technology

Feb 2020 – Dec 2022

University of New South Wales

Sydney, Australia

- Thesis: Efficient Credentials Revocation Using Cryptographic Accumulators
- Supervisor: Prof. Dr. Hey-young Paik
- Award with Excellence

Bachelor of Aviation Management

Feb 2015 – Nov 2017

Massey University

Auckland, New Zealand

PREVIOUS AND CURRENT EMPLOYMENT

Visiting Researcher

August 2026 – February 2027

Carnegie Mellon University

Pittsburgh, PA, USA

- Working with Prof. Dr. Sarah Scheffler and Prof. Dr. Travis Breaux on requirements engineering and ontology for decentralized identity applications.

Researcher

July 2023 – July 2027

University of Zürich

Zürich, Switzerland

- Research focuses on understanding and evaluating decentralized identity systems by examining the relationship between conceptual models and implemented behavior. Studying how responsibilities are distributed within systems, how requirements are specified and assessed, how abstract roles are operationalized, and how systems can be evaluated against their specifications.
- Working on a research project sponsored by the EU Horizon and in collaboration with companies such as STMicroelectronics, Collins Aerospace, and Trust Square. The project is addressing cybersecurity for IoT-enabled environments through a multi-stakeholder sharing of threats and mitigations. The project's result is Artwork.ID system, which offers capabilities to track artworks.
- Planning, developing, and editing of academic publications.

Researcher

Oct 2022 – Dec 2022

University of New South Wales

Sydney, Australia

- Literature review on blockchain opportunities in scientific data management.
- Development of a test environment for data storage with IPFS and Filecoin.
- Smart contract development based on privacy-preserving Oasis blockchain.
- Leading workshop on blockchain potentials for university data management.

TEACHING AND SUPERVISION OF STUDENTS

Teaching Assistant

July 2023 – July 2027

University of Zürich

Zürich, Switzerland

- Teaching and providing support to several university courses, including:
 - * Certificate of Advanced Studies in Blockchain (CAS)
 - * Protocols for Multimedia Communications (PMMK)
 - * Blockchains and Overlay Networks (BCOLN)
- Supervised nine Master (three ongoing) and nine Bachelor students (two ongoing).

Teaching Assistant

May 2021 – Oct 2023

University of New South Wales

Sydney, Australia

- Tutoring and running lab sessions for undergraduate and postgraduate students for:
 - * Data Structures & Algorithms (COMP9024)
 - * Database Systems (COMP9311/ZZEN9311)
 - * Securing Fixed and Wireless Networks (COMP9337)
 - * Cryptocurrency and Distributed Ledger Technologies (COMP6451)
 - * Software Architecture for Blockchain Applications (COMP6452)
 - * Engineering Project Management (GSOE9820)

AWARDS

UZH Doc.Mobility Fellowship

March 2026

ACTIVITIES IN KNOWLEDGE TRANSFER AND OPEN SCIENCE

Peer Reviewer

July 2024 – Present

- ACM Distributed Ledger Technologies Journal
- IEEE World Forum on Public Safety Technology (WF-PST)
- IEEE Workshop on Decentralized Ecosystems and Technologies for Emerging Computing Trends
- IEEE International Conference on Blockchain Computing and Applications
- IEEE Transactions on Network and Service Management
- IEEE Journal of Communications and Networks
- IEEE Journal of Social Computing
- Elsevier Journal of Information Security and Application
- Elsevier Journal of Information Processing & Management

TECHNICAL SKILLS

Languages: JavaScript, HTML/CSS, Solidity, Python

Frameworks: React, Node.js, Truffle, Hardhat, RESTful API

Developer Tools: Git, Jamovi

Libraries: pandas, NumPy, Web3